(Please write your Enrollment Number) Enrollment No. 04801172021

(3)

End-Term Examination
(CBCS)(SUBJECTIVE TYPE) (Offline)
Course Name: B. Tech CSE-AI Semester:VI
(May, 2025)

Subject Code: BAI 306	Subject: Digital Image Processing
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
	a) Explain the concept of gamma correction in image enhancement. How do different values of gamma affect the output image intensity?	
	b) Why median filters are best suited for salt and pepper noise removal?	
	c) Explain the process of conversion of spatial domain image into frequency domain using Fourier transform.	
	d) What is periodic noise in a digital image and how does it affect image quality?	
	e) State differences between Lossy and Lossless image compression.	
	f) What is pseudo color image processing? Discuss its applications in image analysis.	
	g) Explain regional descriptors and boundary descriptors with an example each.	
	h) Explain morphology and different morphological Algorithms.	
	UNIT I	
Q2	a) What is point processing in spatial domain? Discuss any two techniques. b) Given a 3X3 grayscale image: $\begin{bmatrix} 255 & 200 & 100 \\ 100 & 255 & 200 \\ 200 & 100 & 255 \end{bmatrix}$ (i) Perform bit plane slicing and extract 0th and 7th bit plane. (ii) Apply Sobel filter Sx and Sy on (1,1) pixel.	(10)
Q3	a) A 3-bit grayscale image of size 16 × 16 (i.e., MN = 256 pixels) has the following input image histogram. Perform histogram specification to match this image to the desired histogram.	(10)

Input Histogram:

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	4	6	14	22	36	48	60	66

Desired Histogram:

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	0	6	18	28	32	46	58	68

	UNIT II	
Q4	a. Briefly explain the concepts of Inverse Filtering and Wiener Filtering in image restoration with their mathematical formulae. b. An image is degraded in the frequency domain by a blur function H(u,v) and the observed image G(u,v) is:	(10)

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$$G(u,v) = \begin{bmatrix} 90 & 100 & 110 \\ 120 & 130 & 140 \\ 150 & 160 & 170 \end{bmatrix}, \quad H(u,v) = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

Using Inverse Filtering, compute the restored image F (u, v). Show all necessary steps.

Q5	a. Define Image Restoration. Briefly describe the mathematical model of the image degradation/restoration process. b. An observed degraded image G(u,v) blur function H(u,v) and noise power spectrum $S_N(u,v)$ and $S_F(u,v)$ are given in the frequency domain as: $G(u,v) = \begin{bmatrix} 100 & 120 & 140 \\ 160 & 180 & 200 \\ 220 & 240 & 255 \end{bmatrix}$, $H(u,v) = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$ $S_N(u,v) = 10$ and $S_F(u,v) = 1000$. Compute the restored image F(u,v). Show all necessary steps.	(10)
	UNIT III	
Q6	a) Describe the Canny Edge Detection algorithm in detail. Explain each of its main steps and discuss how they contribute to accurate edge detection in images. b) Given an RGB colour with R=51, G=76 and B=25 convert it to: (i) CMYK values (ii) Grayscale value	**(10)**
Q7	a) Explain the concept of redundancy in image data. Describe various types of redundancy found in images. b) A piece of information I is represented by 250 bits using code A, and the same information is represented by 90 bits using another code B. Calculate the compression ratio and the coding redundancy of code A in comparison to code B.	**(10)**
	UNIT IV	
Q8	a) A binary image X and structuring element B are given. Calculate: (i) X^c (ii) $Y1 = X \oplus B$ (iii) $Y2 = X^c \ominus B$ (iv) Opening operation (v) Closing operation	**(10)**

X:

0	1	0	0	1	0	1
0	0	1	1	1	1	1
1	1	1	0	1	0	1
0	0	1	1	0	1	1
1	1	0	1	0	1	0

B:

0	1	0
1	1	1
0	1	0

Q9	a) Compare structural methods with decision-theoretic methods in terms of their approach to classifying patterns. b) Explain the importance of pattern classes in developing a classification system for object recognition.	(10)